

Roll No.

TBT-506(b)

**B. TECH. (BIOTECHNOLOGY) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022**

NETWORK BIOLOGY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Give various definitions of Systems Biology. In brief throw light on associated disciplines of systems biology. (CO1)

OR

- (b) Along with proper diagram elaborate Bus, Ring and Star topology in a computer network. (CO1)

2. (a) Define graph theory in your own words. Throw light on the following terms used in graph theory : (CO1)

- (i) Loop
- (ii) Vertex
- (iii) Edges
- (iv) Points

OR

- (b) Throw light on various sources of PPI data. (CO1, CO2)

3. (a) What is cytoscape ? What are its various features that make it a software of choice for network representation and building ? (CO1)

P. T. O.

(2)

OR

(b) Elaborate various types of network edges along with proper diagram.

(CO2)

4. (a) What is Yeast Two-Hybrid assay ? Outline its various steps with proper diagram.

(CO2)

OR

(b) Elaborate the concept of following, according to graph theory : (CO1)

(i) The degree of a network

(ii) Shortest path

(iii) Transitivity

5. (a) Describe STITCH database and the data contained in it. What are the applications of data in STITCH database ?

(CO1)

OR

(b) Give any five examples of protein-protein interaction in humans along with a proper diagram.

(CO2)